PROCUREMENT PACKAGE:	ME087 – SAG Mill Ball Charger
EPC **PROJECT NO.**	
EQUIPMENT NUMBER(S):	**EQUIPMENT NAME(S):**
F13-BLC-401 F13-CH-474 F13-CH-475	SAG MILL BALL CHARGER SAG MIIL BALL CHARGER FEED CHUTE SAG MILL BALL CHARGER DISCHARGE CHUTE
Description:	The SAG Mill Ball Charger package consists of the following: - Ball Charger Drum, Drum Cover, Removeable Guards and Drive Assembly - Feed Gate Assembly, Isolation Gate, Pneumatic Cylinder and Local Pneumatic Control Panel - Discharge Chute The SAG Mill Ball Charger is required to supply steel balls to the SAG Mill via the SAG Mill Feed Conveyor at a rate determined by the plant control system. The isolation gate is used for isolating the Ball Charger and Feed Chute for maintenance. The SAG Mill Ball Charger Feed Bin (Concrete) has a capacity of 11 to 13 days' consumption of steel balls which are fed to the SAG Mill Ball Charger Feed Bin by CAT 972. The SAG Mill Ball Charger will draw balls from the feed bin via the discharge chute, and on to the SAG Mill Feed Conveyor by the ball feeder (fixed speed) drum. The DCS will activate the ball charge process to feed the SAG Mill as required. The steel balls fed onto the SAG Mill Feed Conveyor will be counted by a ball counting device incorporated in discharge chute of the SAG Mill Ball Charger. Refer to Figure 1 on last page for model screenshot for more details.
Reference Specifications:	SPEC-00-G-05 – Technical Specification for Protective Coating 2000-F00-STS-GE-10001 – Packing and Shipping Specification 2000-F00-STS-GE-10002 – Site Specific Information Specification 2000-F00-STS-EL-10002 – Low Voltage Motor Specification 2000-F00-STS-EL-10008 – Electrical Enclosures Specification 2000-F00-STS-EL-10014 – Preferred Equipment Suppliers for Electrical, Instrumentation and Controls Specification 2000-F00-SPC-ME-10002 - Mechanical Equipment General Specification 2000-MEP-ME087-SCH1 – Vendor Document Schedule

SCOPE OF WORK

Work Included

- Design, supply and fabrication of one (1) SAG Mill Ball Charger including but not limited to:
  - Ball feeder drum including drum liner, drive assembly and bearing assemblies.
  - Inlet and discharge chutes.
  - Gate valve assembly including pneumatic actuator, isolation gate and local pneumatic control panel.
  - Discharge chute.
  - Bolted drum pocket inserts.
  - Ball counting device.
  - Associated instrumentations.
- All ancillary items and special tools normally supplied with or required to adjust and maintain all components associated with the equipment.
- All fasteners, gaskets, packers, shims, and any other items necessary for site assembly.
- Certified lifting lugs and suspension points, clearly stamped with the load limit and test certificates.
- Stainless steel nameplates engraved with manufacturer, model, serial number, and equipment number.
- Surface treatment of proprietary items to manufacturer's highest standard suitable for operation in an outdoor wet process plant environment subjected to hypersaline process and wash-down water.
- Priced recommended spares list.

Specifications

- All equipment to be supplied in accordance with the above referenced specifications.
- Any non-conformances with the specifications shall be clarified with the principal for approval.

Documentation / Quality:

- Vendor data and documentation in accordance with 2000-MEP-ME087-SCH1.
- Shop assembly and testing to the greatest extent possible with accompanying documentation including but not limited to:
    - Dimensional checks.
    - Testing of electric motors, protection, and control devices.
    - Calibration of instruments.
    - Operation of manual, pneumatic or electric mechanisms for valves.
    - Noise testing.
    - Testing of all other auxiliaries to confirm functionality.
- Performance guarantees and testing (state extents).

Shipping:

- Packaging suitable for transport as per 2000-F00-STS-GE-10001.
- Delivery as nominated in the commercial documentation.

Work Excluded

Excluded from this scope of works shall be the following:

- SAG Mill Ball Charger Feed Bin
- Support structure.
- Hold-down bolts, nuts and washers.
- Supply of variable speed drive.
- Supply of all power, control and instrumentation cabling outside battery limits.
- First fill of lubricants.
- Spare parts.
- Offloading the equipment at site.
- Installation at site and commissioning at site

Battery Limits

The works covered by this datasheet extend up to, and include the following battery limits:

- Underside of equipment base plates and bearing assembly.
- Inlet flange of SAG Mill Ball Charger Feed Bin Air inlet connection to isolation gate.
- Electrical terminals of all supplied equipment
    - Electric motor and ball counter terminal boxes

Design Requirements

General

- Equipment will be operating in a harsh outdoor environment; thus, materials and surface treatments must be suitable selected to guarantee the longevity of the components.
- Design and materials of construction to be suitable for the specified duty and environment.
- The ball charger drum shall include standard pockets sized for 150mm steel balls and bolted pocket inserts sized for 125mm steel balls to be used when fitted.

Control System

- Supplier is to provide a detailed control philosophy and control tables for the code to reside in the plant DCS.

Lifting:

- Where practical all assemblies and components shall be provided with suitable lifting lugs or eye bolts to allow for easy off-loading and installation.
- Lifting lugs shall be stamped with a unique ID, the Working Load Limit (WLL) and provided with a Certificate of compliance stating that the lug conforms to relevant Australian Standards.

DUTY SPECIFICATION	UOM	EPC DATA	NOTES
Site Data			
Site Location		Western Australia	1
Site Data		Refer to - 2000-F00-STS-GE-10002 for Site Specific Information	1
Application Data			
Duty		Continuous	1
Location		Outdoors	1
Operating hours per day	hr/day	24	1
Operating days per week	day/week	7	1
Operating weeks per year	weeks/year	52	1
Mechanical Availability	%	>95	1
Design Life	years	30	1
Expected Ambient Operating Temperature Range	°C	-5°C to 50°C	1

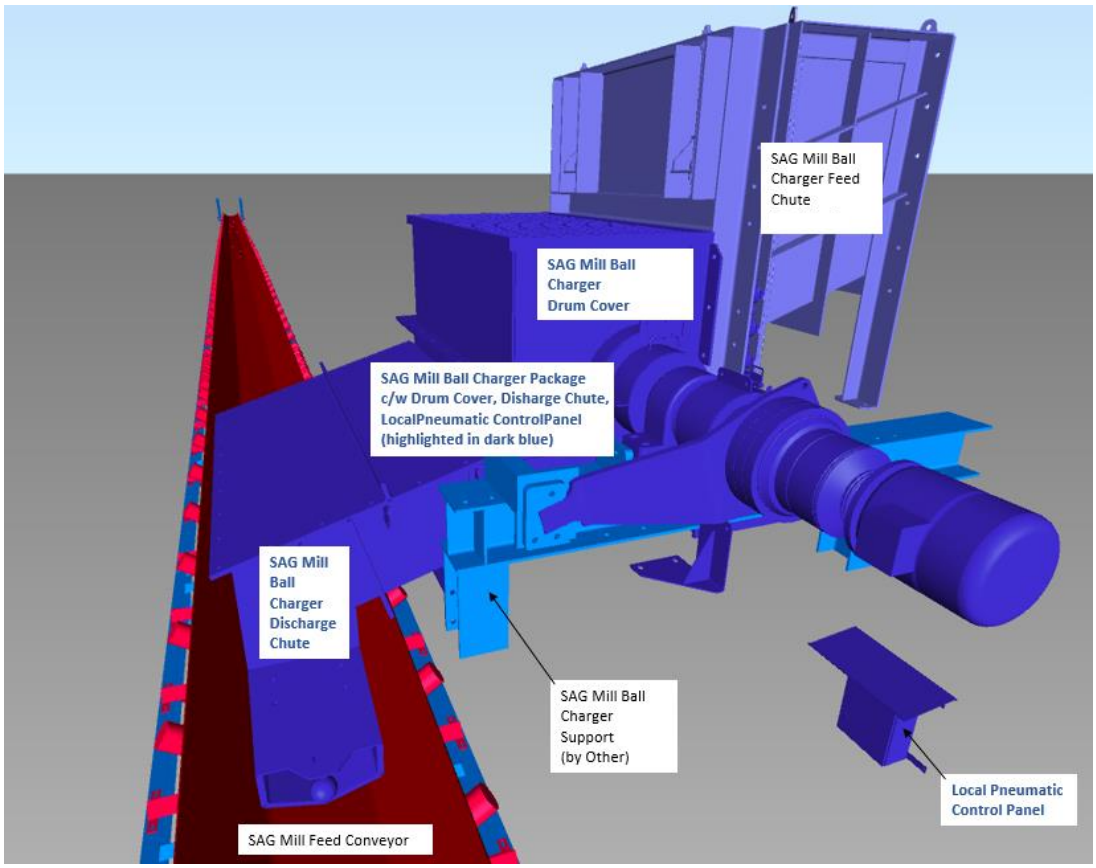
DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA	NOTES
Process Requirements				
Ball Diameter and Material	mm	125 steel balls, Optionally 150 steel balls, Vendor to confirm viability		3
Nominal Ball Mass (each)	kg	8.04		1
Ball Consumption Rate	kg of ball / tonne of mill feed rate	0.40		1
Mill Feed Rate (dry)	tph	1,833 – 2,182		1
Nominal Ball Additional Rate	kg/h	733 - 873		1
Nominal Ball Additional Rate	balls/h	91 - 109		1
Design Ball Additional Rate	balls/h		Up to 2880 balls/h if needed	2
Conveyor Belt Details	-	1500mm wide 35 deg trough angle 2.6m/s belt feed		1
Key Dimensions				
Length	mm	Refer Figure 1 on the last page for some key dimensions		1
Width	mm			1
Height	mm			1
Gear Reducer				
Manufacturer and Model	-		DANA (Brevini)	2
Type	-		Bevel Planetary	2
Ratio	-		355:1	2
Rated Power	kW		5.5	2
Service Factors (At 100% Motor Power)	-		1.53	2
Thermal Rating	kW		6.8	2
Rated Torque	kN.m		20	2
Service Factors (At 100% Motor Power)	-		1.53	2
AGMA Life at Operating Conditions	hr		>200 000	2
AGMA Life at Design Conditions	hr		>200 000	2
Maximum Oil Temperature	° C		80	2
L10 Bearing Life	hr		*74 248	2
Gear Lubrication Details	-		VG220	2
Bearing Lubrication Details	-		VG220	2
High Speed Coupling				
Manufacturer	-		N/A	2
Model	-		N/A	2
Torque Rating	kN.m		N/A	2
Low Speed Coupling				
Manufacturer	-		MAV3008	2
Model	-		175X300	2
Torque Rating	kN.m		44.7	2
Ball Feeder Drum				
Drum Diameter	mm		510	2
Drum Shaft Speed	rpm		4	2
Drum Material of Construction	-		500NB SCH 80 pipe	2
Drum Shaft Material of Construction	-		1045	2
Bearing Manufacturer	-		NSK / FSQ or SKF	2

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA	NOTES
Bearing Housing Part Number	-		FSSNFD530TACN	2
Bearing Type and Part Number	-		SN 500 series	2
Bearing L_{10} Life	hr		>60000	2
Motor Details				
Motor Manufacturer / Type	-	Teco MAXe3 (Preferred)	Yes	2
Absorbed / Installed Power	kW		2.8/5.5	2
Motor IEC Frame Size / Motor Mounting Type	-		132M/B5	2
Number of Poles / Full Load Speed	# / rpm		4	2
Power Supply (Voltage / Frequency / Phase)	V / Hz / ϕ	415 / 50 / 3		1
Enclosure (IP) Rating	-	IP66	IP66	2
Starter Type (VSD/DOL/Soft Start)	-	DOL		1
Motor Rated for VSD Control (Yes/No)	-	No		1
Hazardous Area Rated (Yes, Rating/No)	-	No		1
Temperature Protection (Yes/No)	-	No	No	3
Anti-Condensation Heater (Yes/No)	-	Yes	No (not required for this application)	3
Surface Protection/Paint Spec	-		Factory standard	2
Shut Off Gate Actuator				
Power Source	-	Pneumatic	Pneumatic	3
Pneumatic Actuator Manufacturer	-		SMC	2
Pneumatic Actuator	-		SMC	2
Bore	mm		125	2
Stroke	mm		280	2
Instrument Air Pressured Required	kPag		600 to 700	2
Instrumentation Provided (No Auto Switches & Type / Other)	-		Proximity Switch	2
Isolation Gate Material of Construction	mm		Mild Steel	2
Ball Counting Device				
Manufacturer / Model	-		P&F OBD300 Proximity Sensor	2
Output Signals (Provide details)	-		4-20mA	
Local Pneumatic Control Panel				
Manufacturer / Model	-		SMC	2
Enclosure Material of Construction	-	316SS	316SS	3
Enclosure IP Rating	-	IP65	IP65	3
Control Voltage	-	24VDC		1
No of Digital Inputs / Outputs Provided	-		N/A	2
Locally Mounter Indication and Gauges	-		N/A	2
Inlet Process Connection Size			TBA	2
Outlet Process Connection Size		Push in fittings suitable for 12mm OD nylon tube	TBA	3
Local Ball Counter Control Panel				
Manufacturer / Make / Model	-		N/A (OBD300 will be wired to a JB mounted on machine)	2
Materials of Construction	-	316SS		1
IP Rating	-	IP65		1
Control Voltage (DC)	V	24	24	3
Installed / Consumed Power	kW		N/A	2
Voltage / Frequency / Phase	V / Hz / ϕ	240 / 50 / 1	N/A	3
Status Signals - List Details	-	Ready/Running/Fault/E-Stop	N/A	3
Type of Contacts	-	Dry	N/A	3
Control Signals - List Details	-	Remote reset	N/A	3
Platework				
Thickness	mm		Varies (typically 10mm)	2
Platework Material of Construction	-		Mild Steel	2
Discharge Chute Liner Material / Thickness	mm		Linatex 40 or equivalent / 15mm	2
Surface Protection				
Preparation	-		RCR System 1	2
Primer type and DFT	µm		Zinc Rich Epoxy 50micron	2
Top-coat type, colour and DFT	µm		High Build High Solids Epoxy 150micron + Paracryl IF540 50micron	2

DUTY SPECIFICATION	UOM	EPC DATA	SUPPLIER DATA	NOTES
Weights				
Motor	kg		113	2
Gearbox	kg		300	2
Drum/Shaft Assembly	kg		1400	2
Maximum Lifting Weight for Transportation / Installation	kg		5100	2
Heaviest Item for Maintenance	kg		1400	2
Shipping Data				
Total Shipping Weight	kg		5100	2
Largest Item Shipping Weight	kg		5100	2
Largest Component Shipping Size (L x W x H)	mm		2900 x 3100 x 2500	2
Extent of Pre-Assembly for Shipping - Describe	-		Fully assembled except pneumatic cabinet	2

Notes:

1. Data supplied by EPC The Supplier shall comply with this data.
2. Data shall be provided by Supplier.
3. Data proposed by EPC The Supplier shall confirm data or state alternative.



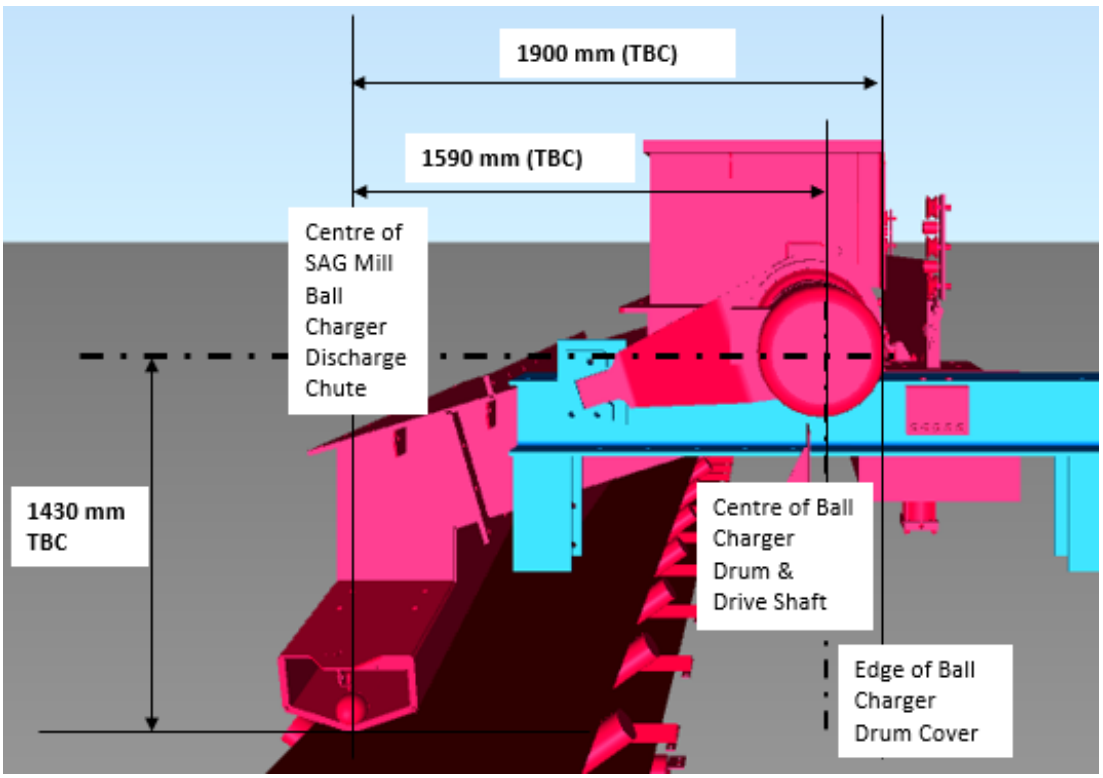


Figure 1 – Model Screenshots